Independent reproduction check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Independent reproduction check before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Peer review and response

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Writing the methods section

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Robustness across conditions

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

Complete the supported Independent reproduction check trace and record the first step you can explain without help.

Use W2: Trace the worked case for Independent reproduction check. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Independent reproduction check explanation against the supplied facts before discussing it.

Use the worked case for Independent reproduction check. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Reproducer uses different validation split because no split IDs were provided. Diagnose.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Independent reproduction check facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 Trace the worked case

> Trace the worked case. Apply the supplied Independent reproduction check facts and keep your reasoning.

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Independent reproduction check facts and keep your reasoning.

A report omits this qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Independent reproduction check facts and keep your reasoning.

Reproducer uses different validation split because no split IDs were provided. Diagnose.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Independent reproduction check facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Independent reproduction check to Reproducible AI Research Study, identify a limit and choose the next useful test.

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Reproducer uses different validation split because no split IDs were provided. Diagnose.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Independent reproduction check

> An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score. Explain the labelled step before checking the changed condition.

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

Original correct count is 5/6. Rerun is 4/6. Case Q3 is the only scoring disagreement. No new model **output**s are generated.

AI ACADEMY / AI-7/33

Independent reproduction check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Compare exact artefacts

Verify data and rubric versions before changing the result.

2. Locate disagreement

The score difference is concentrated in Q3.

3. Resolve by the rubric

Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

CHANGE ONE THING

Q3's saved file is missing from the archive.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

1. Resolve by the rubric

2. Compare exact artefacts

3. Locate disagreement

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Resolve by the rubric Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Q3′s saved file is missing from the archive. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. A missing record prevents checking the contested decision.

Repair a missing building block

Review Peer review and response (ai-7/32). Goal: Explain the **mechanism** and terminology of peer review and response. Short practice: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

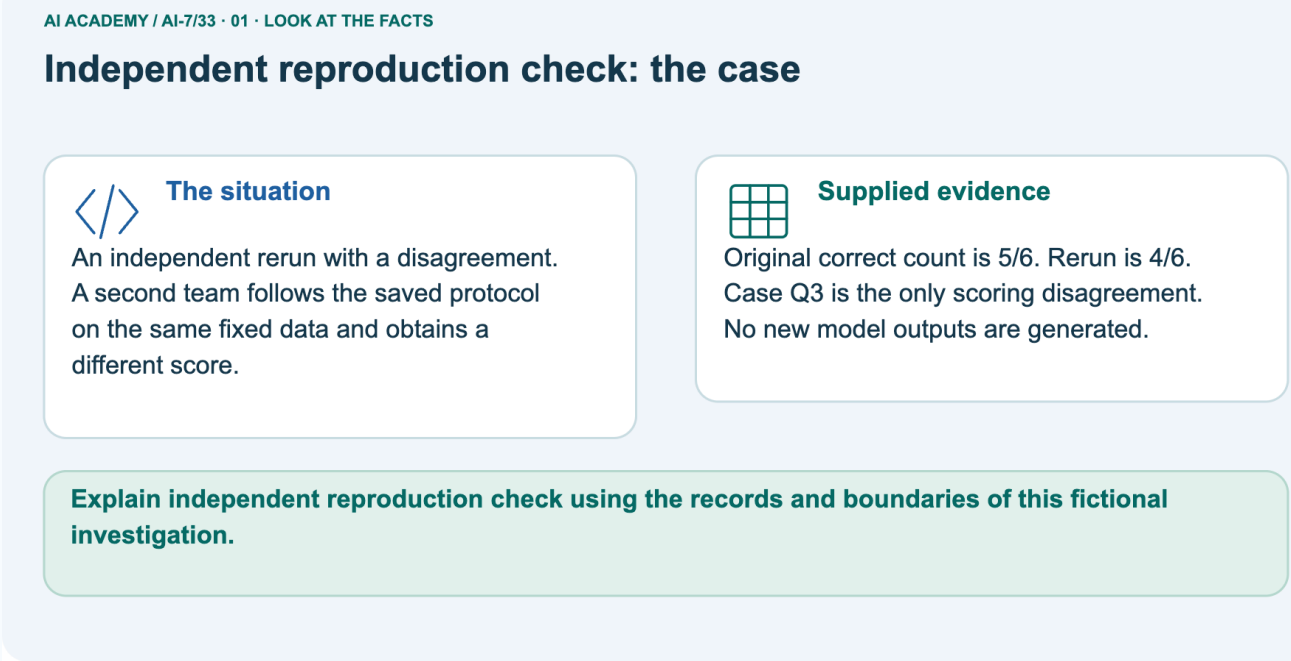
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Independent reproduction check picture from its labels, then record a relevant limit.

01. Independent reproduction check: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

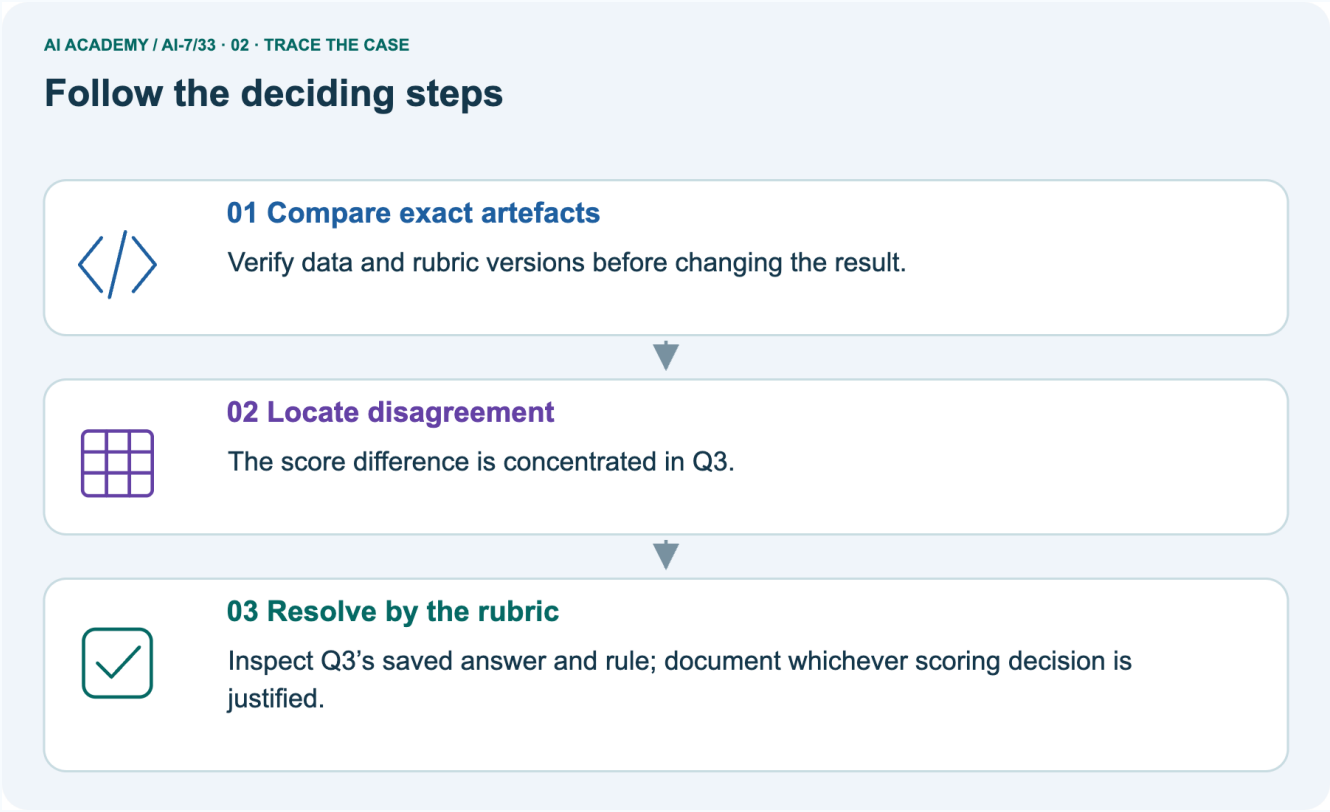
An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score. Supplied facts: Original correct count is 5/6. Rerun is 4/6.

Case Q3 is the only scoring disagreement. No new model **output**s are generated.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



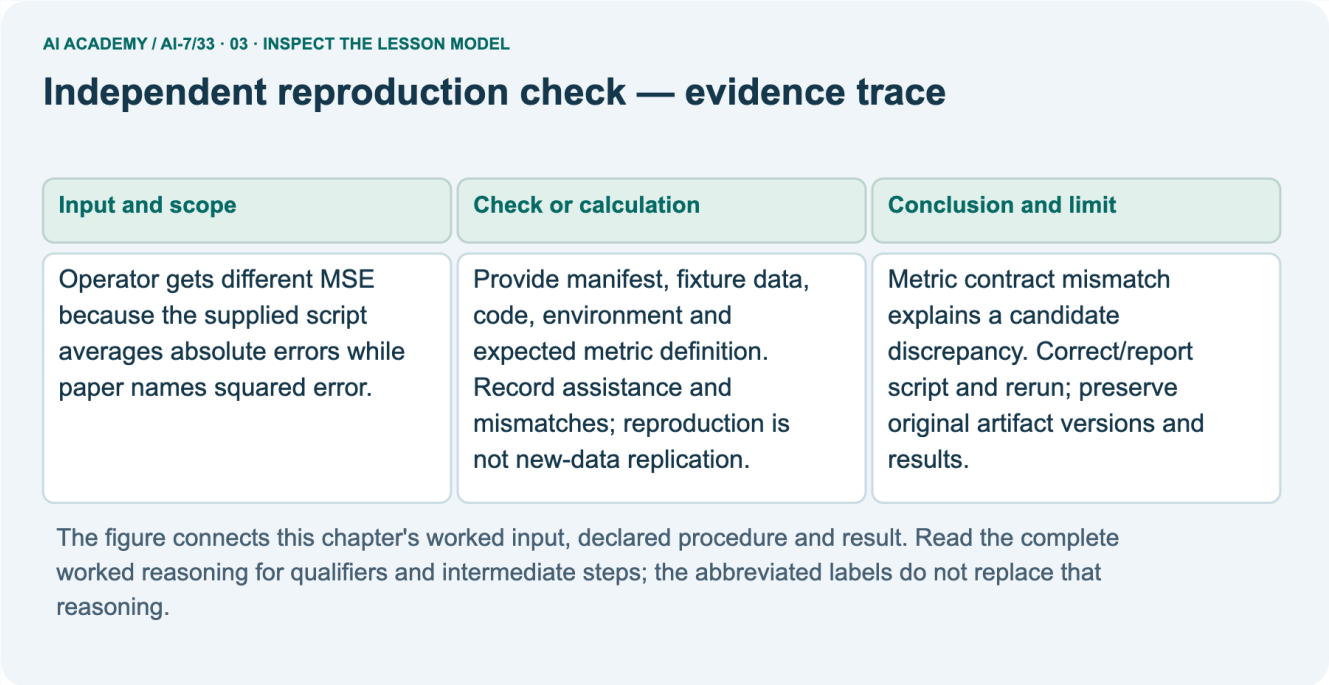
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare exact artefacts: Verify data and rubric versions before changing the result. → Locate disagreement: The score difference is concentrated in Q3. → Resolve by the rubric: Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Independent reproduction check — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Independent reproduction check — **evidence** trace.

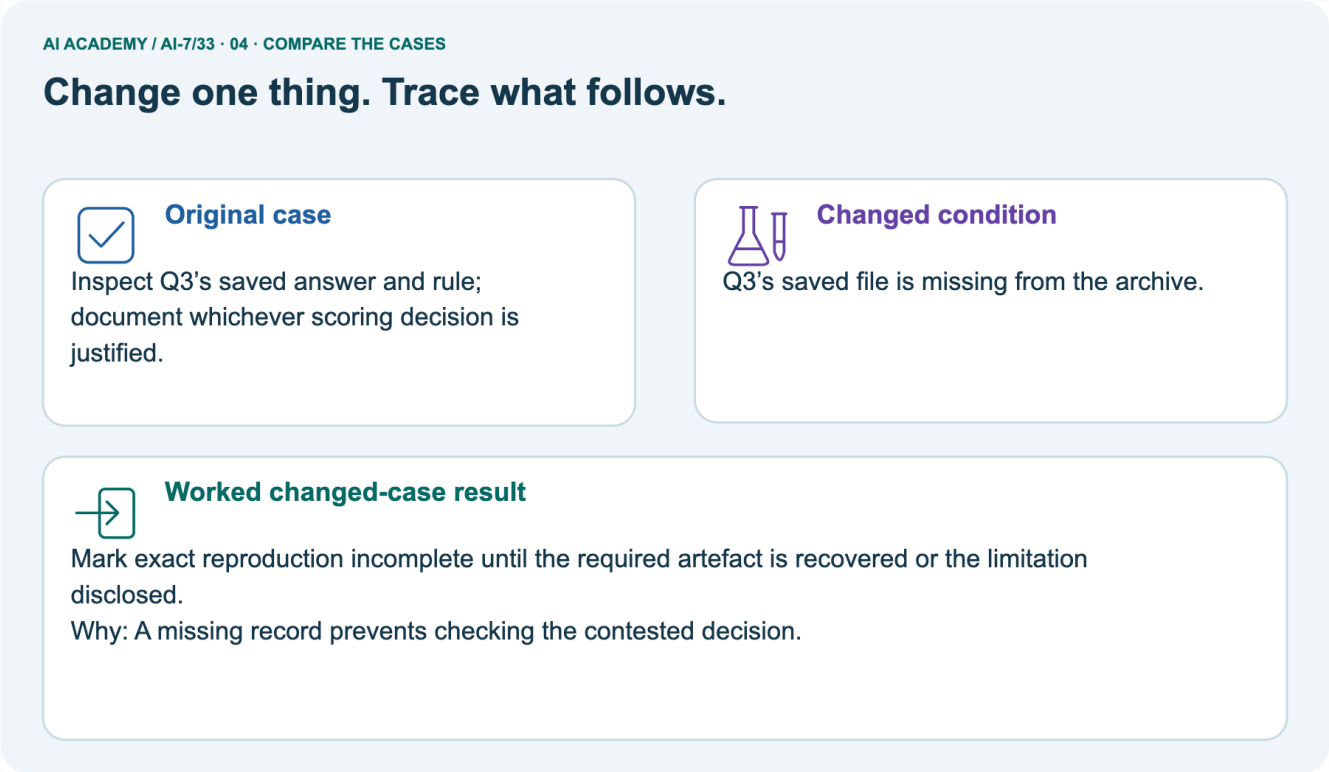
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. | Provide manifest, fixture data, code, environment and expected metric definition.

Record assistance and mismatches; reproduction is not new-data replication. | Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

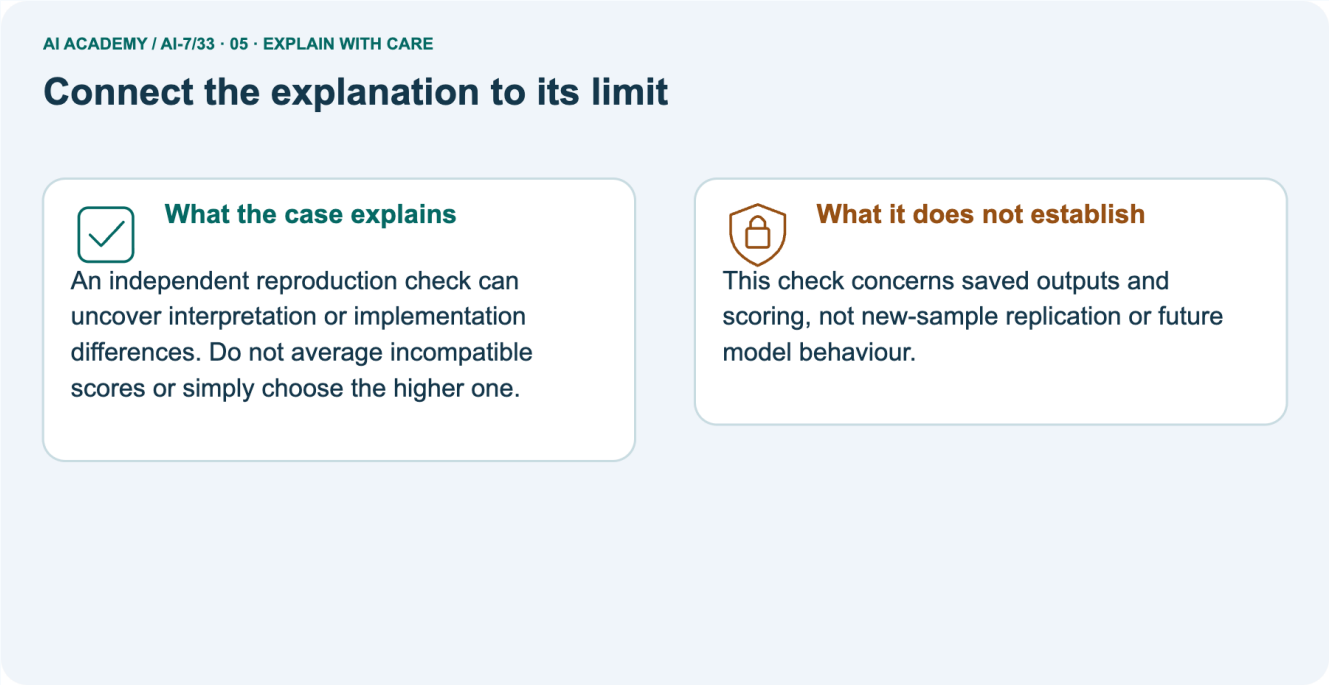
Original case: Inspect Q3's saved answer and rule; document whichever scoring decision is justified. Changed condition: Q3's saved file is missing from the archive.

Expected result: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. Reason: A missing record prevents checking the contested decision.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

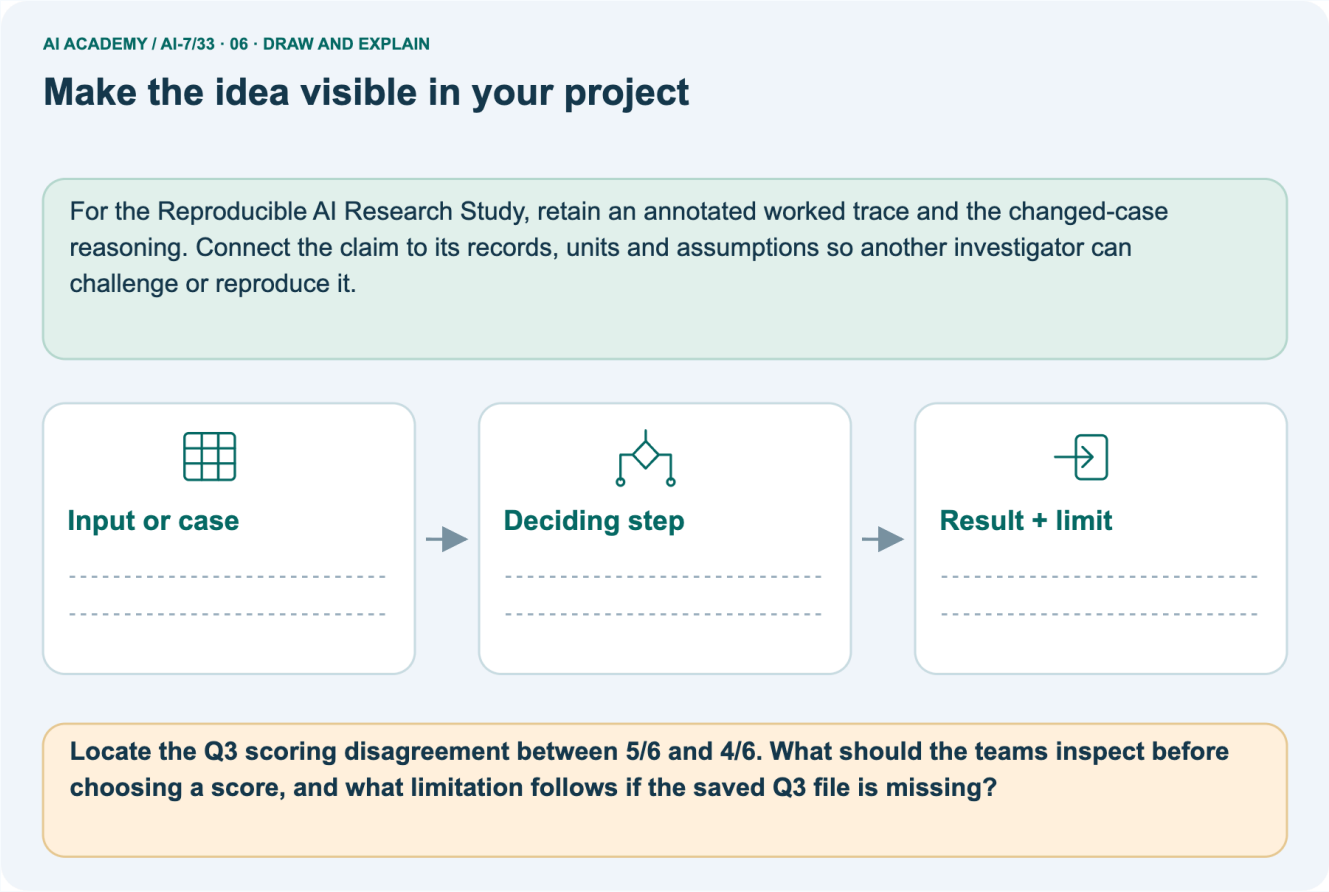
An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Limit: This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.